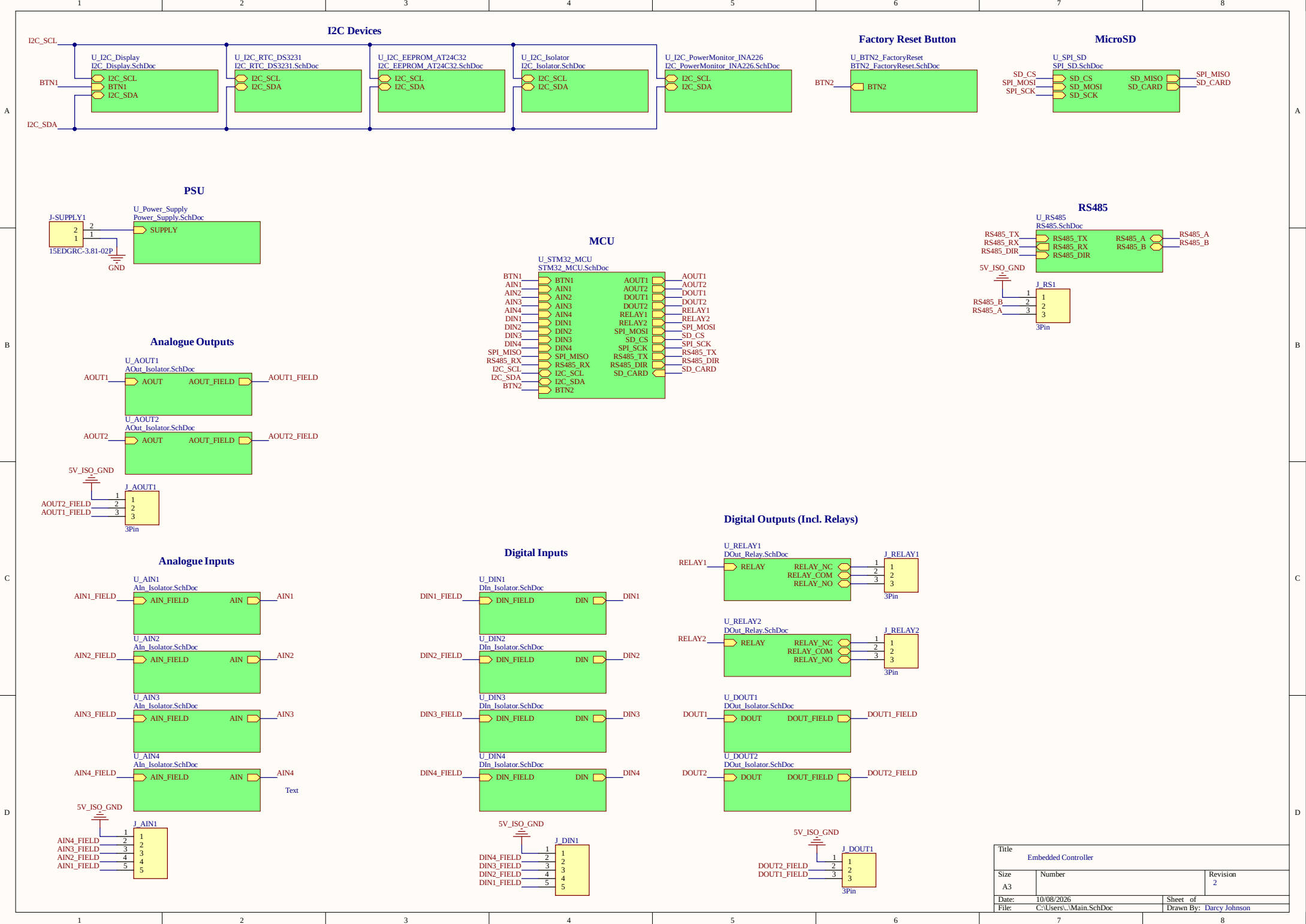
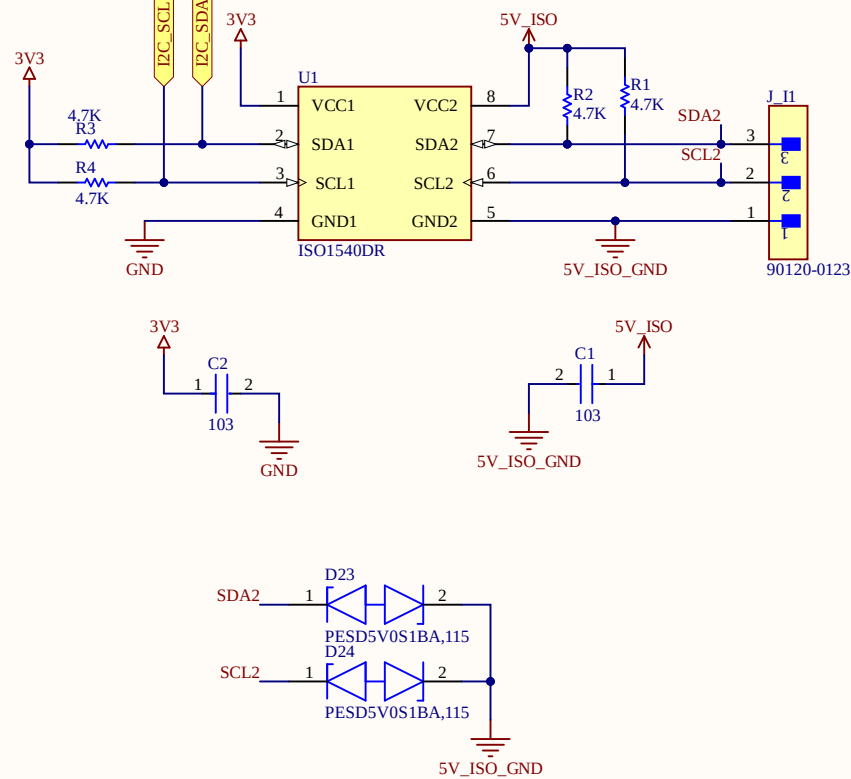




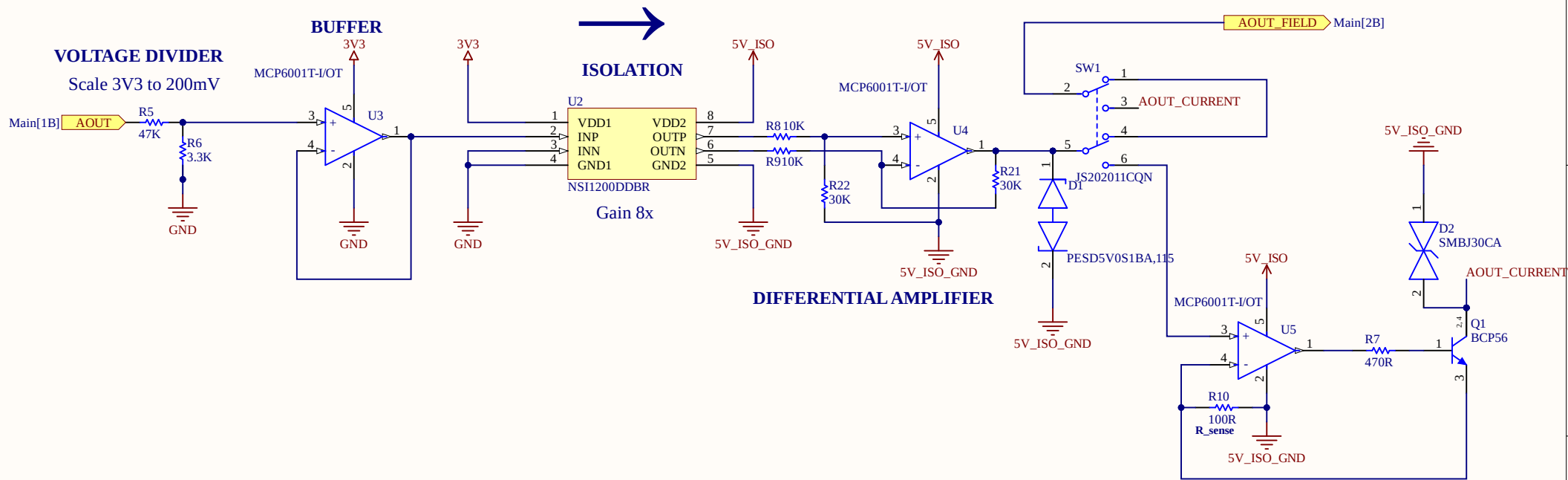
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Size	Number	Revision
A3		2
Date:	10/08/2026	Sheet of
File:	C:\Users\Main\SchDoc	Drawn By: Darcy Johnson



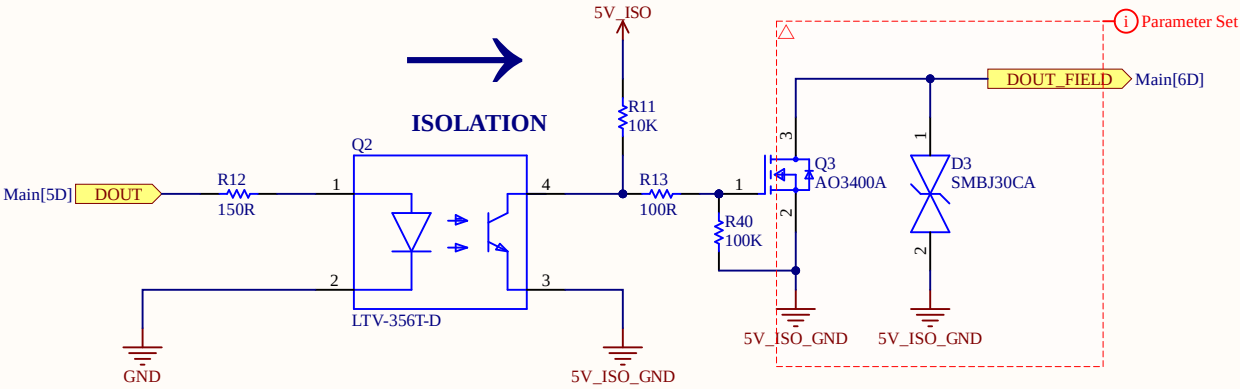
ISO1540 SDA, SCL Pull ups required - floating is invalid input.

Hence, SDA, SCL are 5V only. Field user must use external level shifter for 3.3V sensors (see PCA9306)

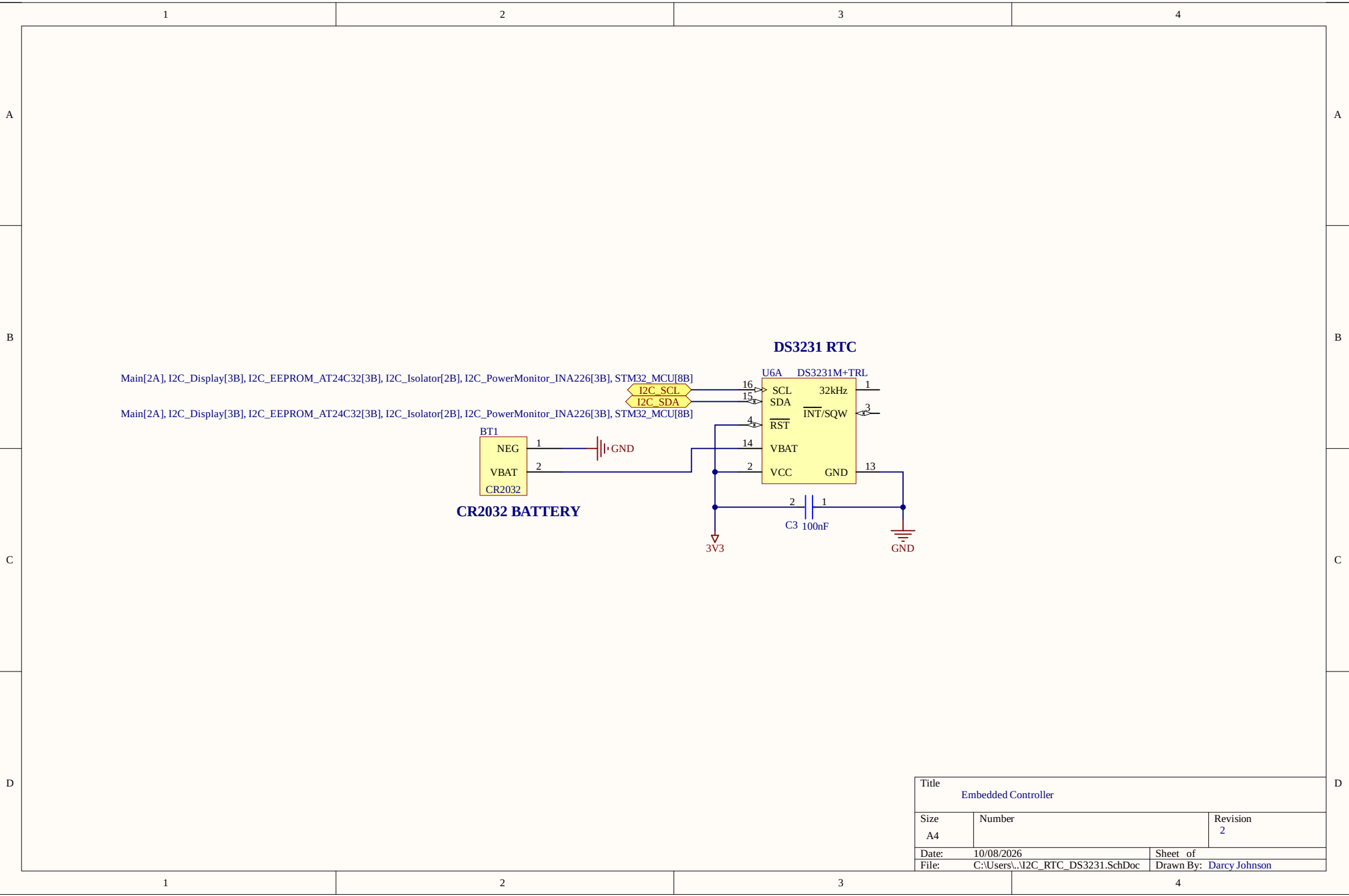
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Date:	10/08/2026	Sheet of
File:	C:\Users\...I2C_Isolator.SchDoc	Drawn By: Darcy Johnson

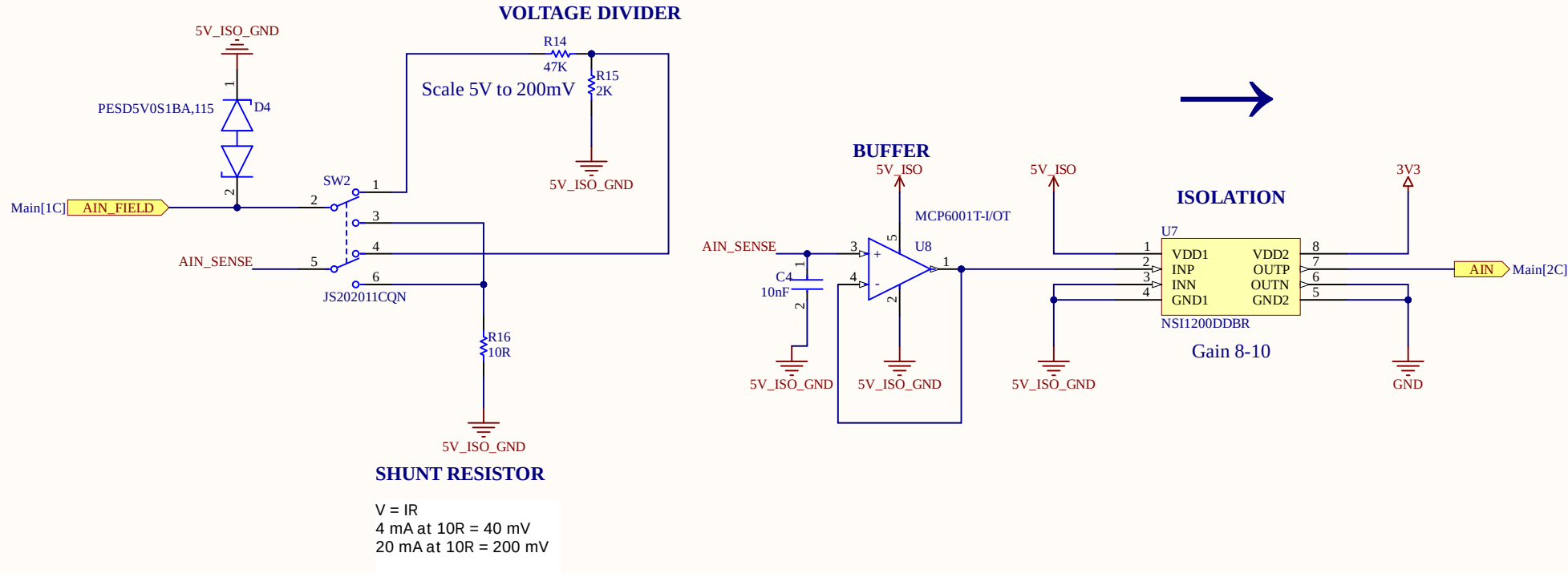


Title		
Embedded Controller		
Size	Number	Revision
A4		2
Date:	10/08/2026	Sheet of
File:	C:\Users\...\AOut_Isolator.SchDoc	Drawn By: Darcy Johnson

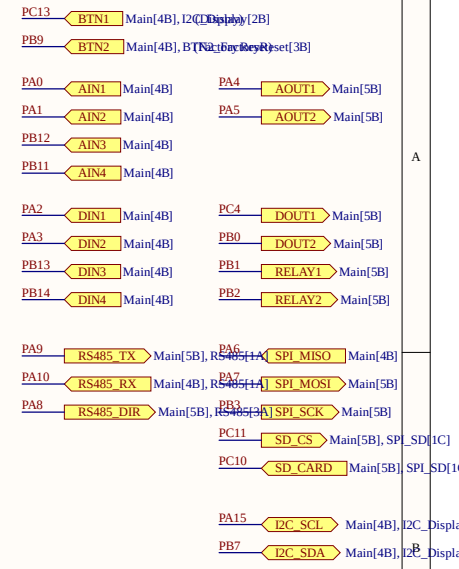
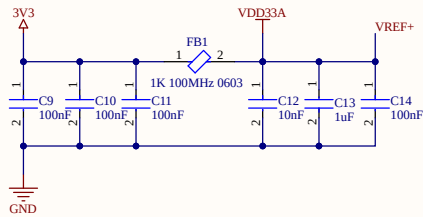
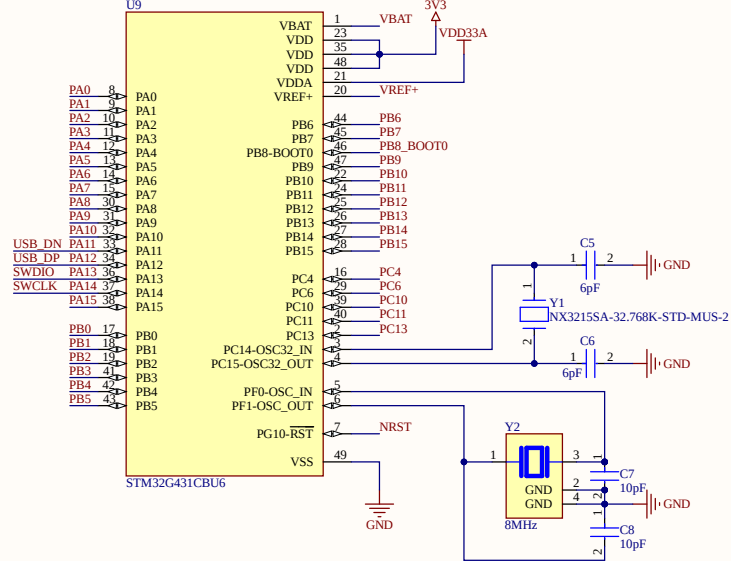
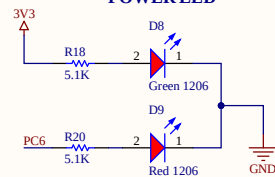
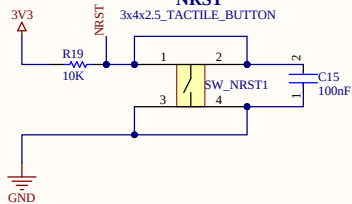
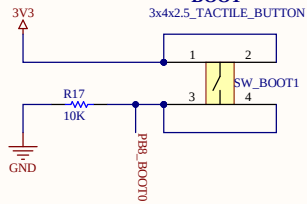
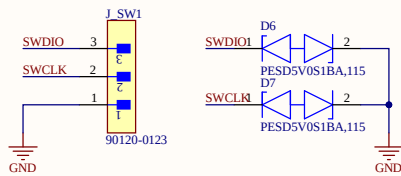
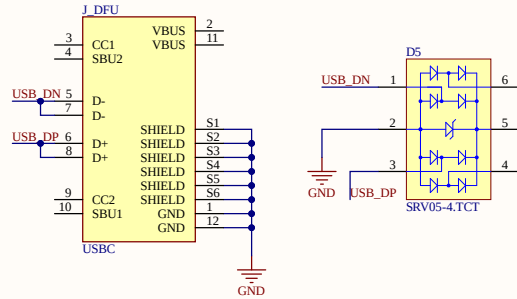


Title Embedded Controller		
Size A4	Number	Revision 2
Date:	10/08/2026	Sheet of
File:	C:\Users\...\DOut_Isolator.SchDoc	Drawn By: Darcy Johnson

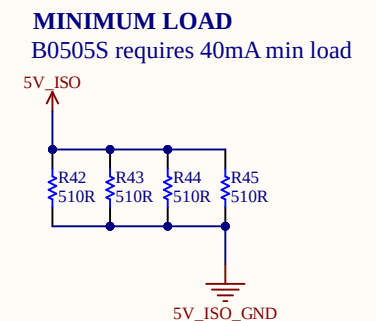
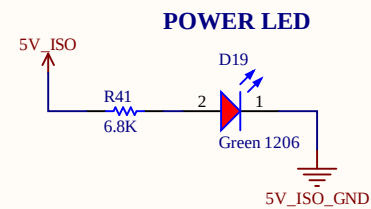
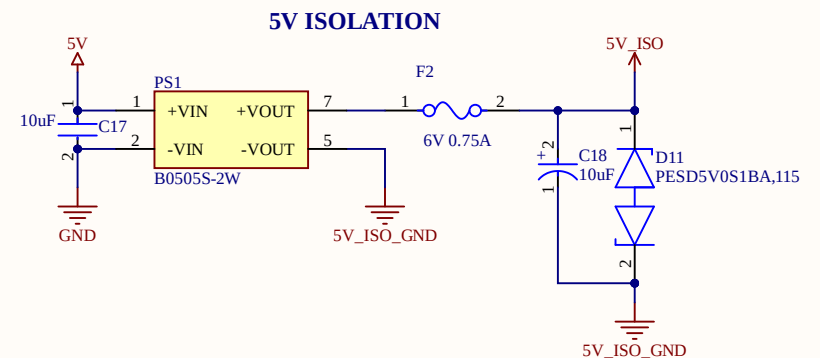
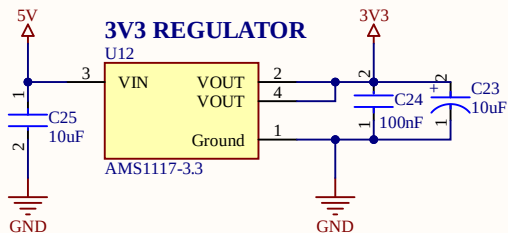
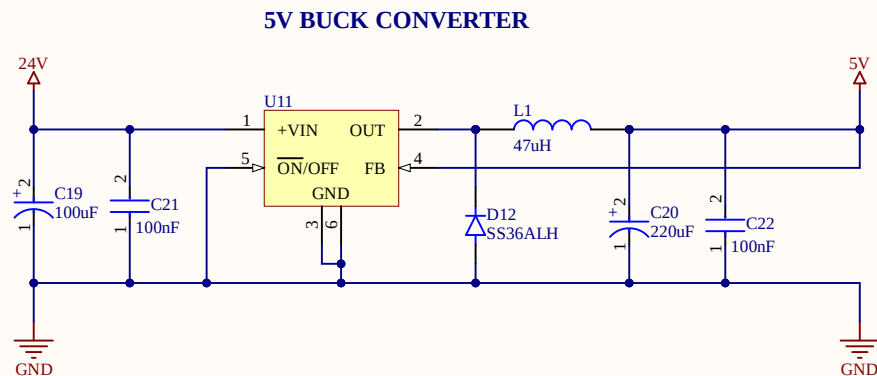
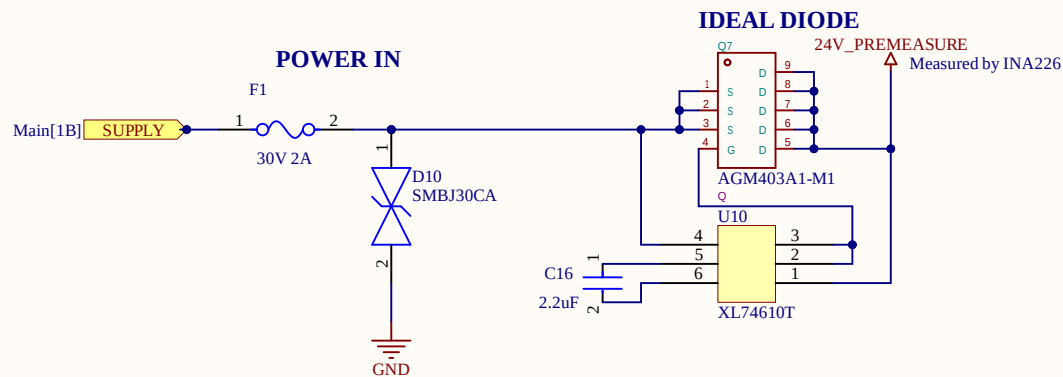




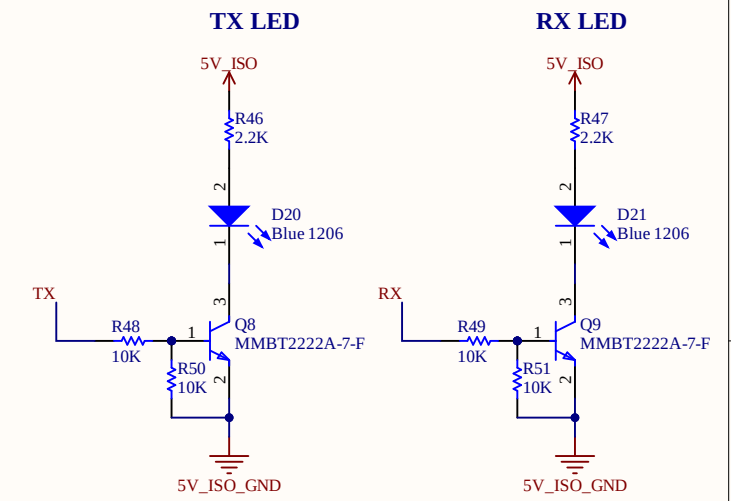
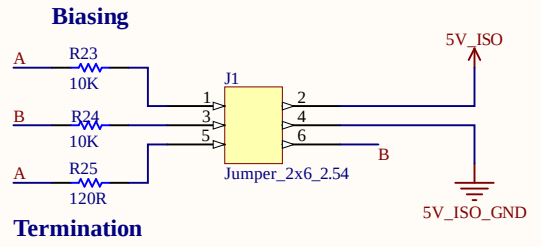
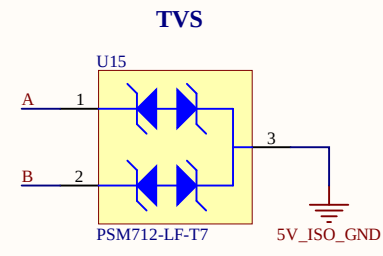
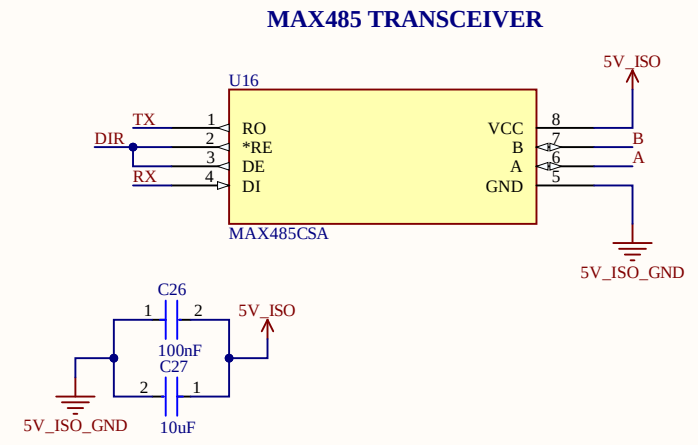
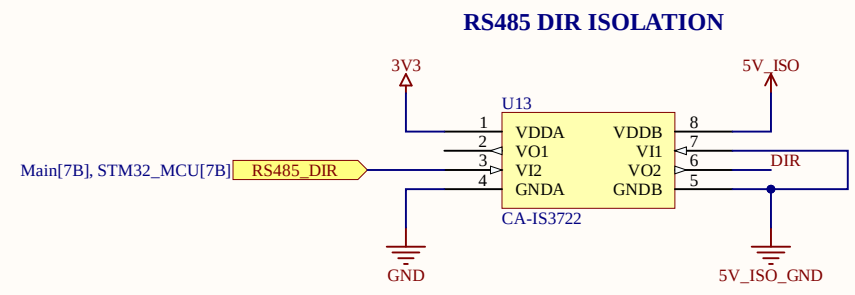
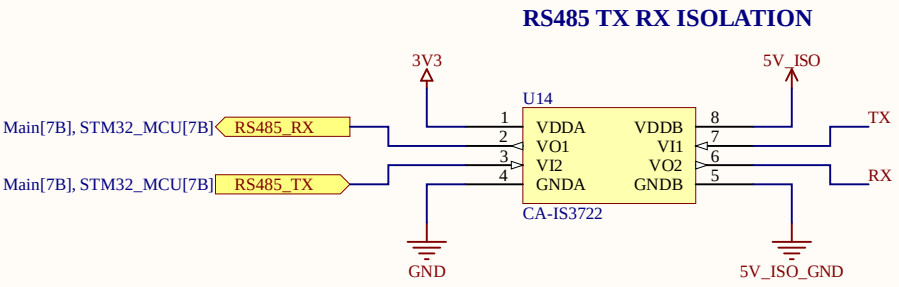
Title		
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Size	Number	Revision
A4		2
Date:	10/08/2026	Sheet of
File:	C:\Users\...\AIN_Isolator.SchDoc	Drawn By: Darcy Johnson



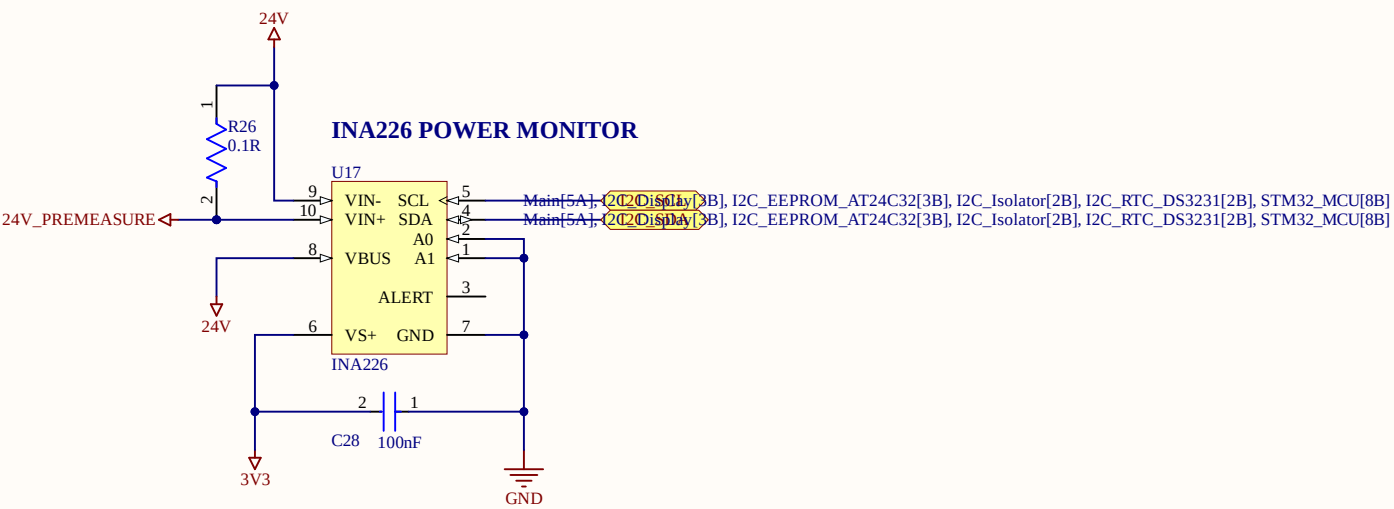
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Size A3	Number		Revision 2
Date: File:	10/08/2026 C:\Users\...\STM32_MCU.SchDoc	Sheet of	Drawn By: Darcy Johnson



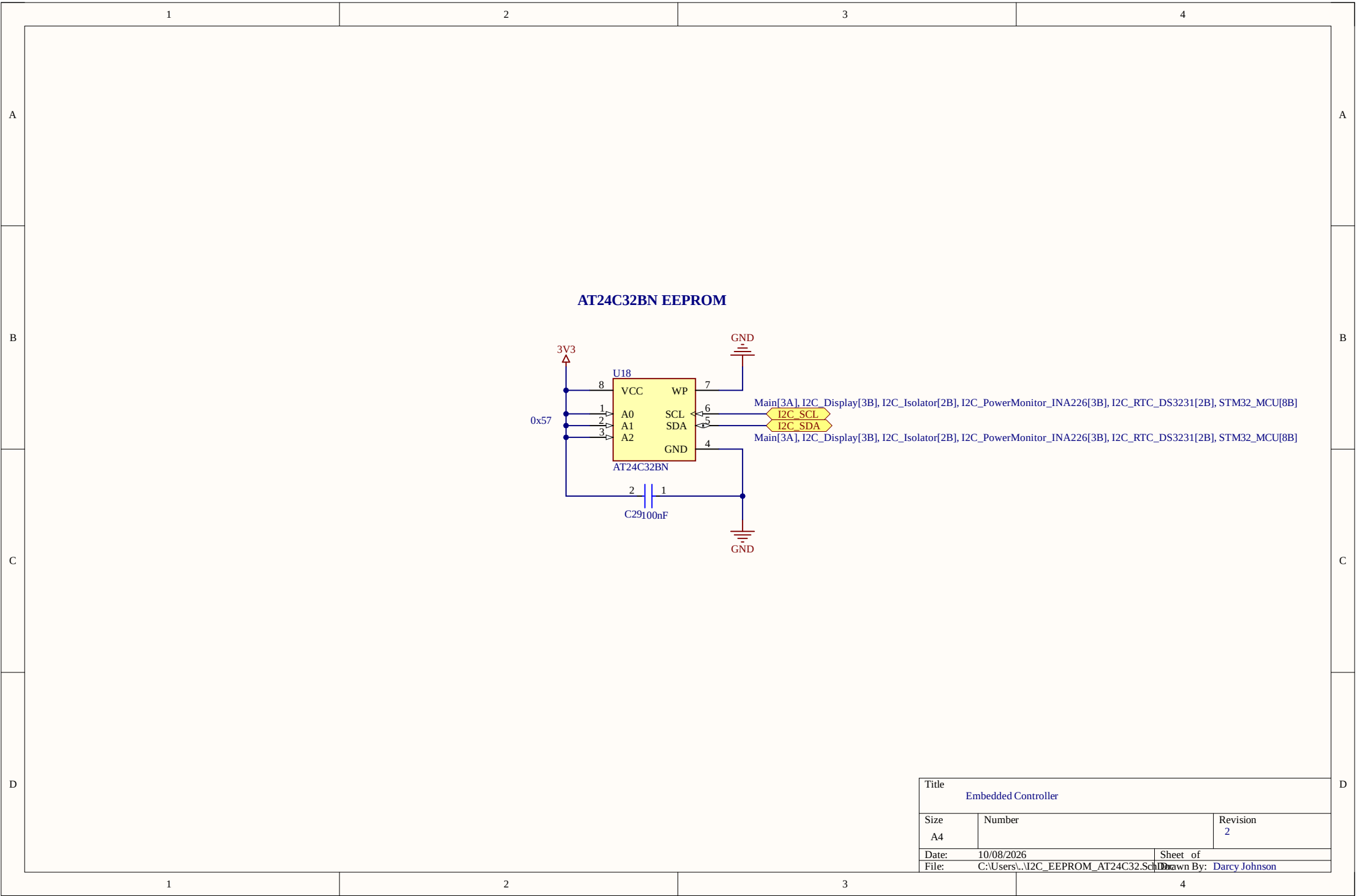
Title		
Embedded Controller		
Size	Number	Revision
A4		2
Date:	10/08/2026	Sheet of
File:	C:\Users\...\Power_Supply.SchDoc	Drawn By: Darcy Johnson



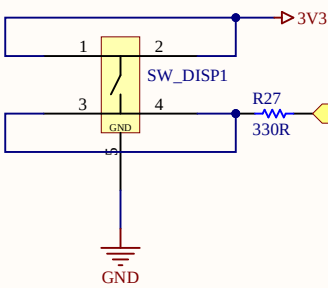
Title		
Embedded Controller		
Size	Number	Revision
A4		2
Date:	10/08/2026	Sheet of
File:	C:\Users\...\RS485.SchDoc	Drawn By: Darcy Johnson



Title		
Embedded Controller		
Size	Number	Revision
A4		2
Date:	10/08/2026	Sheet of
File:	C:\Users\I2C_PowerMonitor_INA226_SchDoc	Drawn By: Darcy Johnson



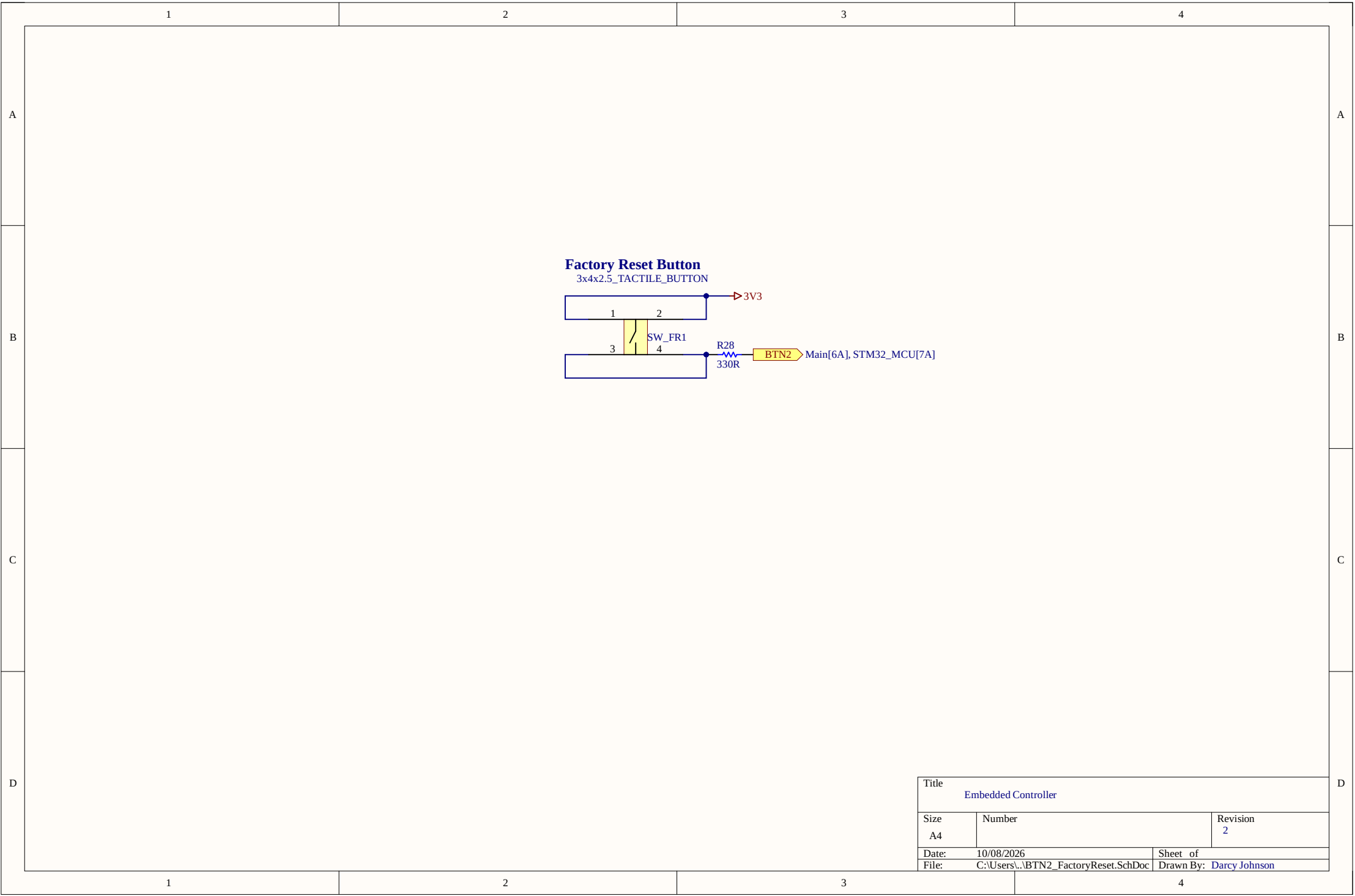
Input Button
6x6x3.1_TACTILE_BUTTON



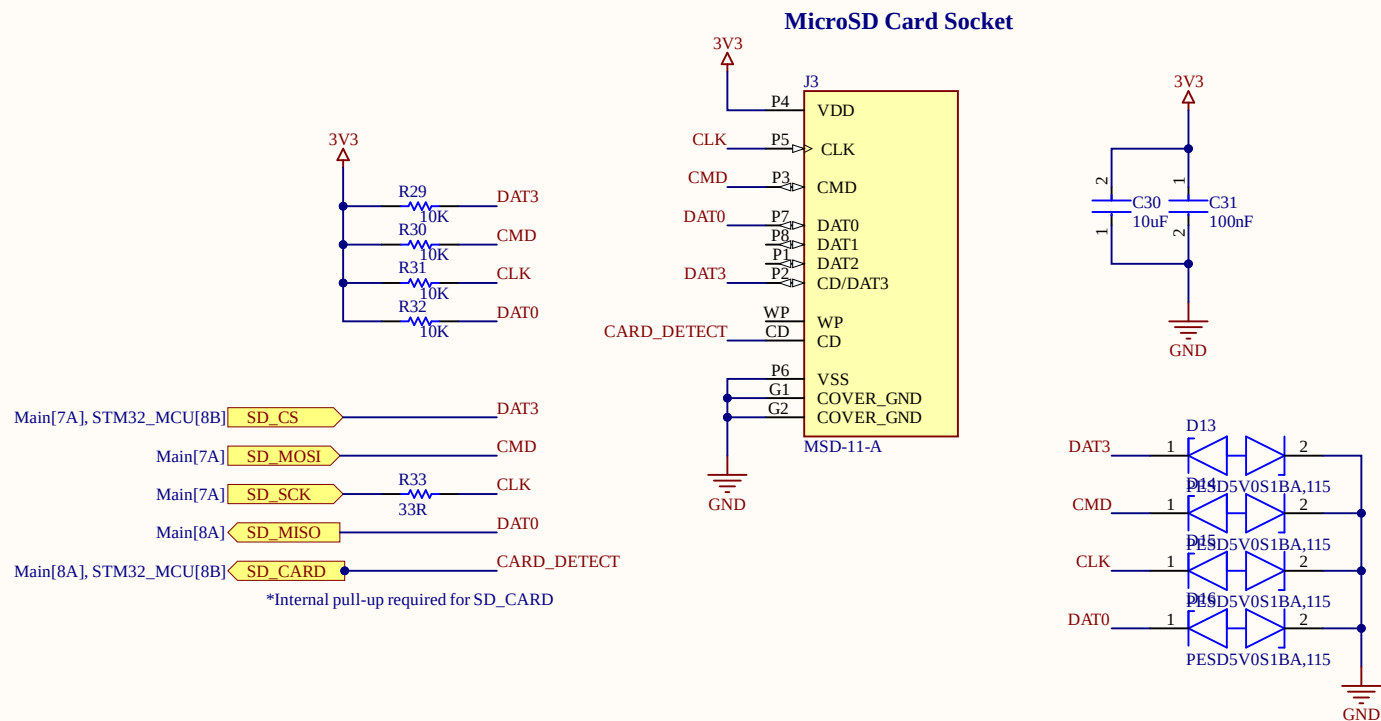
Display Module



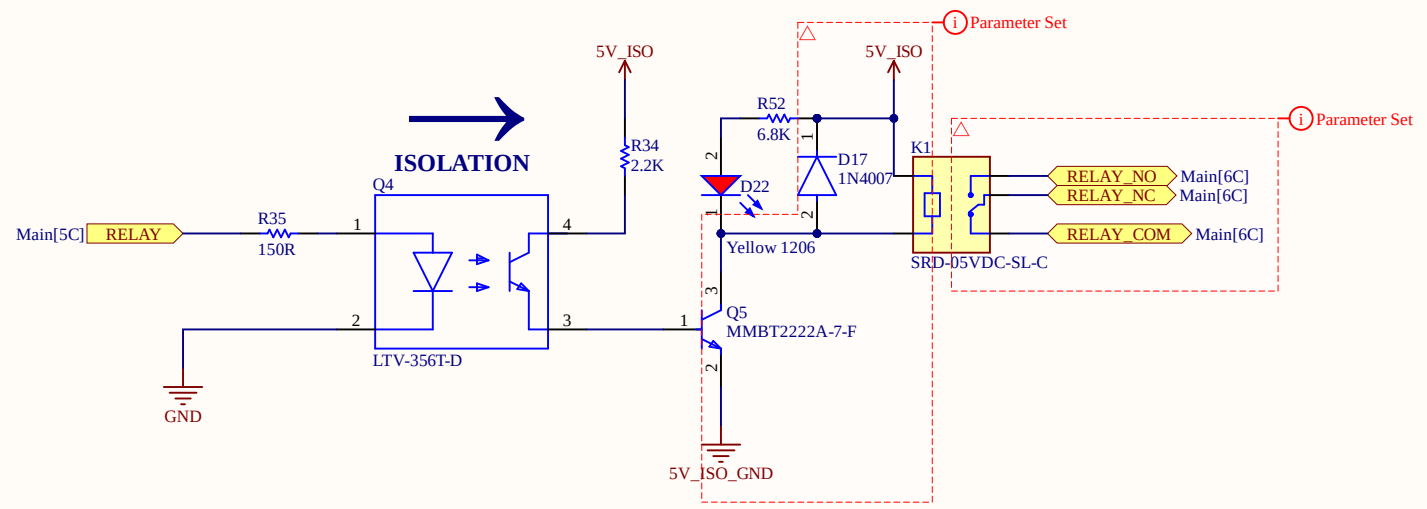
Title Embedded Controller		
Size A4	Number	Revision 2
Date:	10/08/2026	Sheet of
File:	C:\Users\...\I2C_Display.SchDoc	Drawn By: Darcy Johnson



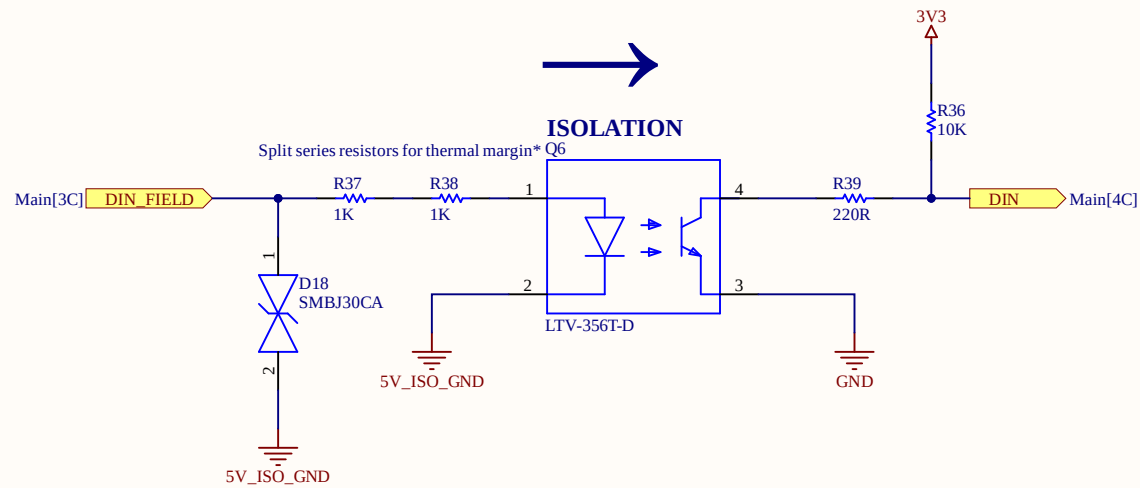
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Size A4	Number	Revision 2
Date:	10/08/2026	Sheet of
File:	C:\Users\...\BTN2_FactoryReset.SchDoc	Drawn By: Darcy Johnson



Title		
Embedded Controller		
Size	Number	Revision
A4		2
Date:	10/08/2026	Sheet of
File:	C:\Users\...\SPI_SD.SchDoc	Drawn By: Darcy Johnson



Title		
Embedded Controller		
Size	Number	Revision
A4		2
Date:	10/08/2026	Sheet of
File:	C:\Users\...\DOut_Relay.SchDoc	Drawn By: Darcy Johnson



*24V input - 1.2V forward LTV356 = 22.8V

$I = V / R$, $I = 22.8 / 2000 = 11.4\text{mA}$

$P = I^2 * R$, $P = 0.0114^2 * 2000 = 0.260\text{ W}$

1x 1206 can do 0.250 W so split evenly across two for 0.125 W each

Title Embedded Controller		
Size A4	Number	Revision 2
Date:	10/08/2026	Sheet of
File:	C:\Users\...\DIn_Isolator.SchDoc	Drawn By: Darcy Johnson